



Identification of Damage in Planar Truss Structures Within a Multiphysics Framework Using Metaheuristic Optimization Algorithms

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Abstract: Accurate damage identification in planar truss structures is essential for ensuring structural reliability, operational safety, and long-term serviceability in complex multiphysics environments. In this study, a structural damage identification framework was developed by integrating metaheuristic optimization algorithms with the finite element method. The particle swarm optimization (PSO) and the genetic algorithm were employed to identify both the location and severity of structural damage. The natural frequencies of the structure were adopted as objective indicators. The optimization process was designed to minimize the difference between measured and computed frequency responses, thereby enabling the precise localization and quantification of damage in individual elements. Furthermore, the applicability of the developed framework to structural systems operating under coupled multiphysics effects was emphasized, thereby enhancing its practical relevance for real-world engineering applications. The proposed approach provides an effective and computationally efficient strategy for structural health monitoring and damage assessment of planar truss systems, with significant potential for integration into intelligent maintenance and reliability management frameworks.

Keywords: Planar truss; Particle swarm optimization; Genetic algorithm; Damage; Natural frequencies

1 Introduction

The truss structure is now an essential part of many different types of structures. Numerous areas, including civil infrastructure, automotive manufacturing, etc., have made extensive use of it. Due to fatigue, aging, environmental effects, etc., truss structures may reduce the level of security service substantially, which could result in accidents in the future. Therefore, damage identification is crucial for preserving the structure's safety and integrity as well as for lowering maintenance costs. It is essential to develop effective and appropriate methods for evaluating truss structure damage, which is why numerous academics from across the world have attempted to address this issue. Figure 1 illustrates Vietnam's self-elevating offshore drilling rig system, highlighting the importance of strength in operation under a multiphysics environment.



Figure 1. Vietnam's self-elevating offshore drilling rig

The advancement of computer technology and software in recent decades has made it possible for us to complete many computationally demanding tasks quickly. Modern math theory and artificial intelligence are becoming more and more popular, and they have been shown to be successful in the majority of sectors. Metaheuristic optimization algorithms have several advantages. For example, they are simple, easy to apply, and are used widely. In order to reduce the processing time of structural damage identification problems, Farhadi et al. [1] suggested a two-stage damage detection approach for damage localization and extent quantification. A residual force vector was used in the first step of the suggested framework to identify potential damage spots. The model-updating technique then evaluated the damage extent of previously detected elements in the second stage using three different optimization algorithms: enhanced colliding body optimization, differential evolution, and particle swarm optimization (PSO). The sine-cosine algorithm, which is another metaheuristic algorithm, has been proven to be very competitive compared with other metaheuristic algorithms and attracted significant attention from researchers in different fields. There hasn't been any research done on applying the sine-cosine algorithm to the truss structure. In order to tackle the truss optimization problem, Zhou et al. [2] improved the sine-cosine algorithm in a number of ways. In order to better align the exploration region with the features of the real optimization scenario, a nonlinear conversion parameter was first set in place of the linear conversion value. Second, the algorithm's global search capability was improved by using the Lévy flight. Thirdly, to improve the local search capability, an elite guidance technique utilizing memory data was suggested. Ultimately, the candidate was chosen using a greedy selection process.

Contreras-Bejarano and Villalba-Morales [3] investigated bio-inspired numerical methods as a substitute for truss structural design optimization. The algorithm procedure was set up for five aspects: (i) the mutation operator; (ii) the inclusion of multi-modal techniques; (iii) the inclusion of parameter control techniques; (iv) the definition of the initial population; and (v) local search heuristics. The differential evolution algorithm proved to be simple to implement. In order to increase the performance of the typical bat method for optimizing engineering problems and truss structure design, Vu-Huu et al. [4] took advantage of another improved metaheuristic approach. The bat algorithm was simple to implement, as was the case with many metaheuristic algorithms, and such a simple approach might handle a large range of issues with significant flexibility. An enhanced grey wolf optimizer algorithm was presented by Sang-To et al. [5]. In terms of the leader wolf's movement direction and a unique parameter that enabled the speedier wolves to prey on places, the improved version was intriguing and complementary. The alpha, beta, and delta wolves used the Lévy flying as a unique navigation method. To handle the local search issue, the leader wolf had a potent weapon. To speed up the algorithm's convergence, a new principle that depicts the omega wolf's hunting behavior was also introduced.

The primary goal of Jawad's work [6] was to use a discrete novel nature-inspired optimization algorithm to produce a more suitable design for truss structures. To get the intended outcomes, the dragonfly algorithm was used. The method was inspired by the static and dynamic swarming behaviors of dragonflies in the natural world. The dragonfly algorithm was first created for issues involving continuous optimization. To address discrete functions of optimization problems and enhance the algorithm's performance, certain modifications were suggested. By locating the cost-effective areas in structural design, optimization techniques aid in reducing the weight of steel structures. The minimization of weight design was searched for using metaheuristic methods. Finding the best design for truss structures in a way that would lower their cost was the primary goal of the study conducted by Saravanan et al. [7]. The truss system's cost-effective portions were found using the most recent enhanced Cuckoo Search optimization technique. Large-scale global optimization issues were solved by Sang-To et al. [8] using a novel shrimp and goby association search technique. Thirteen benchmark high-dimensional functions, ten classical benchmark functions, and a number of practical engineering applications were used to evaluate the algorithm's performance.

Based on an examination of how optimization problems with a weak influence of the global optimal solution impact metaheuristic optimization methods, Su et al. [9] presented a unique particle contribution evaluation mechanism. The particle contribution evaluation method was novel in that it employed deep learning models to determine if a particle was a high-contribution particle inside the influence region of the global optimum based on the feature information, in contrast to the existing mechanisms in this field. This gave metaheuristics extra crucial data from outside the optimization process to direct the particle population's proper evolution. Tian et al. [10] proposed a novel nature-inspired metaheuristic algorithm, i.e., the snow geese algorithm. It was inspired by the migratory behavior of snow geese and emulates the distinctive "herringbone" and "straight line" shaped flight patterns observed during their migration and so on, as further listed with other algorithms [11–13]. Deng [14] proposed the baboon optimization algorithm, a novel metaheuristic algorithm inspired by the hierarchical social structure, foraging techniques, and stress response mechanisms of baboon populations. Through adaptive foraging and stress-induced perturbation processes, the baboon optimization algorithm separated the population into leader, adult, and juvenile layers, allowing for a dynamic balance between local exploitation and global exploration. The baboon optimization algorithm repeatedly demonstrated significant exploitation capacity on unimodal functions. Nevertheless, the baboon optimization algorithm's computational efficiency was limited, and its overall running time occasionally surpassed that of other algorithms in use.

Sheikhi [15] proposed the painted wolf optimization algorithm, another nature-inspired metaheuristic optimization technique. The algorithm was primarily inspired by the hunting tactics and group behavior of painted wolves, who are often referred to as African wild dogs in the wild. This is especially true of their distinctive consensus-based voting rally mechanism, which is fundamentally different from the social dynamics of grey wolves. The creative method involved pack members searching for prey in various locations before holding a pre-hunting vote rally based on the alpha member to decide who would start the hunt and assault the prey. Although metaheuristic algorithms have shown to be quite successful, their resilience is still limited by enduring issues such as early convergence, insufficient exploration, and performance loss in high-dimensional or nonconvex search environments. Saad et al. [16] developed the Fourier transform optimizer, a metaheuristic framework that integrated frequency-domain analysis using the discrete Fourier transform, to address these problems. With the goal of achieving a more successful balance between exploration and exploitation, this approach made it possible for solution updates to be driven by frequency-based transformations. To improve search dynamics, the Fourier transform optimizer combined a number of improvement techniques, including orthogonal learning, differential frequency mixing, Lévy flight perturbations, and Runge-Kutta-inspired updating strategies.

In the study by Wang and Shang [17], a metaheuristic algorithm known as the "traffic jam optimizer" was put forth to address real-world engineering issues. It was inspired by the traffic police officers' directives to drivers during a traffic bottleneck, which ultimately resulted in a smooth traffic scenario. The algorithm was broken down into three stages: first, drivers' autonomous driving behavior caused traffic jams; second, drivers gradually became aware of the traffic situation and changed their driving direction by self-regulation; and third, the traffic police officers involved in directing the traffic, and drivers obeyed their orders and entered the enforced phase. Xu [18] introduced the crocodile ambush optimization algorithm, a metaheuristic algorithm motivated by crocodiles' energy-saving and ambush hunting techniques. To attain a fair trade-off between exploration and exploitation, among other things, the crocodile ambush optimization algorithm included threshold-based solution reinitialization, adaptive energy decay modeling, stochastic leader selection, and so on.

The current study focuses on presenting the damage identification of planar truss structures using two well-known metaheuristic optimization algorithms: the genetic algorithm (GA) [19] and PSO [20]. The former is frequently used to produce high-quality solutions to optimization and search problems via biologically inspired operators like selection, crossover, and mutation. The latter is an optimization algorithm inspired by the social behavior of bird flocking. The traditional finite element method is used to establish the stiffness matrix and mass matrix to determine the natural frequencies, which are adopted as an objective function for optimization. The formulas related to the calculation of natural frequencies can be found in related literature [21–25]. Two truss systems with 21 bars and 54 bars are analyzed to verify the effectiveness of the above metaheuristic optimization algorithms. Finally, conclusions are presented.

2 Methodology

To anticipate the location and degree of damage to truss structures, the finite element method is coupled with the GA and PSO. These optimization techniques can identify and quantify the degree of damage in structural elements by analyzing the difference between actual damage and the frequencies obtained from numerical analysis.

The GA is widely acknowledged as a key technique in the domains of artificial intelligence and optimization. This algorithm is used to address various issues in everyday life activities, from politics to science and from agriculture to industry, and it replicates the theory of evolution [19]. Three principles—selection, crossover, and mutation—describe the GA. Furthermore, an algorithm that mimics swarm behavior is called PSO.

The procedure of the PSO comprises two main sections, i.e., velocity (V) and location (X), for each agent as follows:

$$V_j^d(i+1) = \omega V_j^d(i) + c_1 \times \text{rand} \times [p_j^d(i) - X_j^d(i)] + c_2 \times \text{rand} \times [g^d(i) - X_j^d(i)] \quad (1)$$

$$X_j^d(i+1) = X_j^d(i) + V_j^d(i+1) \quad (2)$$

where, ω , c_1 and c_2 are the weight factors, and p and g are the best locations of the member and swarm at the current time, as shown in Figure 2 [20]. Figure 3a clearly illustrates the operative process of the PSO algorithm. Then these two algorithms are used to predict the damage of the two-dimensional truss structures, respectively.

An objective function is defined as the difference in natural frequencies between the finite element model and actual measurements. The assessment of the PSO and the GA uses the change in the stiffness of the structure. Thereby, severity and location of damage can be identified. In order to use these algorithms, the difference in natural frequencies between the actual damage without constraints and the finite element model is the applied objective function. The definition of the damages is shown in Eq. (3), where a decrease in the bar's modulus of elasticity is

crucial:

$$E_r = (1 - r)E, \quad 0 \leq r \leq 1 \quad (3)$$

where, r is the variable, showing each bar's degree of damage. By minimizing the following objective function, it is therefore simple to determine the position and extent of the harm or damages:

$$O^f = \sqrt{\sum_{i=1}^n \frac{(f_i^a - f_i^e)^2}{(f_i^a)^2}} \quad (4)$$

where, a and e stand for the actual and finite element and n is the number of frequencies taken into consideration. The framework of the PSO/GA is also shown in Figure 3b.

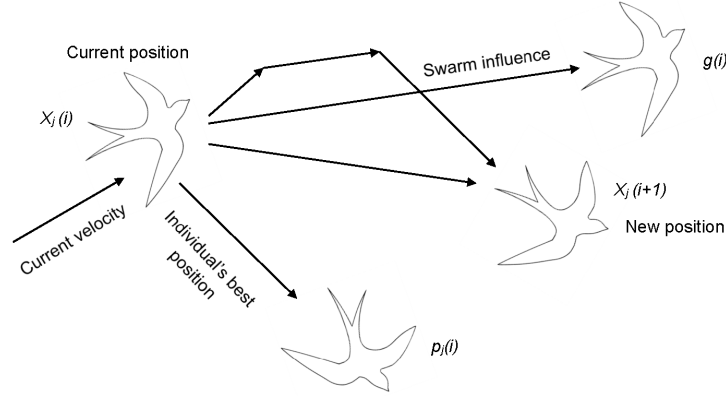


Figure 2. Details for the particle swarm optimization (PSO) algorithm

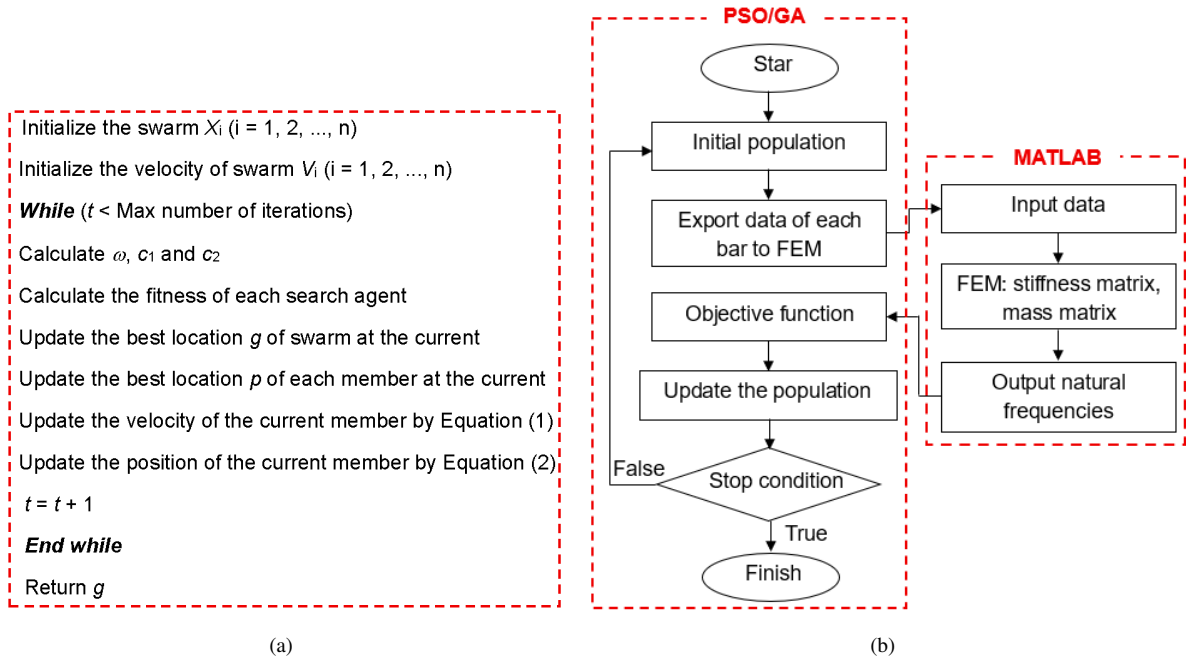


Figure 3. (a) Pseudocode of the particle swarm optimization (PSO); (b) Framework of the PSO/genetic algorithm (GA)

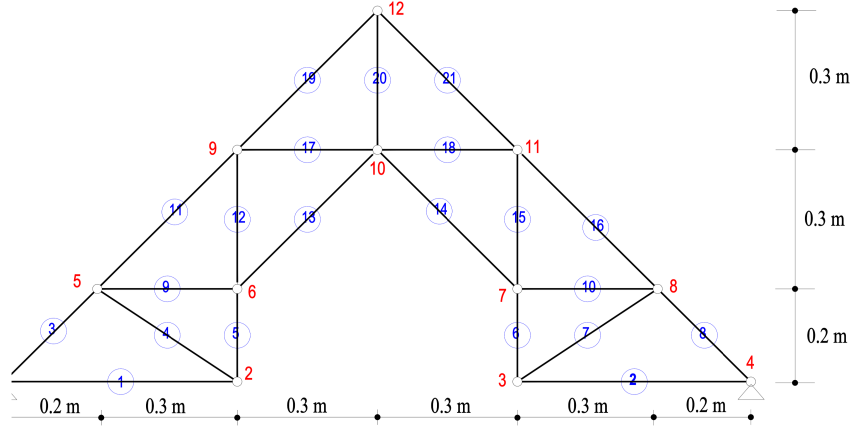
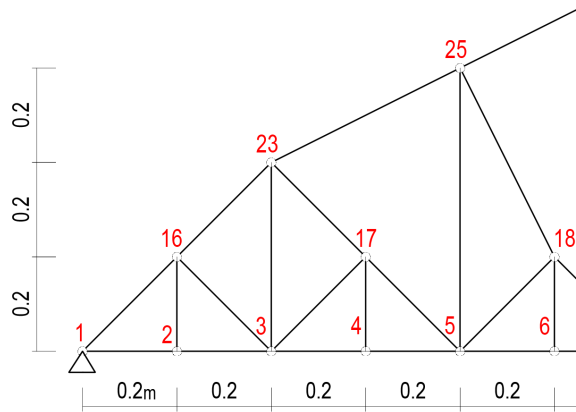
Note: PSO = Particle Swarm Optimization; GA = Genetic Algorithm; FEM = Finite Element Method.

3 Results and Discussion

Two plane truss systems with 21 and 54 bars are discussed in this section. The parameters required for the analysis process are also shown in Table 1. In addition, the geometric shapes and data are illustrated in Figures 4 and 5 and Tables 2–4.

Table 1. Truss properties

Parameter	Symbol	Value
Young's modulus	E	205 GPa
Density of mass	ρ	7850 kg/m ³
Cross-sectional area	A	7.0686×10^{-4} m ²

**Figure 4.** 21-bar truss**Figure 5.** 54-bar truss**Table 2.** Node coordinates of the 21-bar truss

Node	X (m)	Y (m)	Node	X (m)	Y (m)
1	0	0	7	1.1	0.2
2	0.5	0	8	1.4	0.2
3	1.1	0	9	0.5	0.5
4	1.6	0	10	0.8	0.5
5	0.2	0.2	11	1.1	0.5
6	0.5	0.2	12	0.8	0.8

To check the accuracy, some damage scenarios are given in Tables 5 and 6.

The findings show that both approaches—the GA and PSO—produce the intended outcomes for damage structure prediction (for the 21-bar and 54-bar trusses), which are detailed in Figures 6–10. It is evident that both algorithms can achieve the intended outcomes in under 80 iterations for the 21-bar truss and 280 iterations for the 54-bar truss. Furthermore, in most circumstances, the GA's convergence curves outperform those of the PSO (Figures 6b, 7b, and 9b). Both algorithms do a good job of locating the damage bar for global search, although it should be noted that the GA consistently outperforms the particle swarm optimization. The GA yields more competitive results for

calculating the damage in all cases when compared to particle swarm optimization.

Table 3. Node coordinates of the 54-bar truss

Node	X (m)	Y (m)	Node	X (m)	Y (m)
1	0	0	15	2.8	0
2	0.2	0	16	0.2	0.2
3	0.4	0	17	0.6	0.2
4	0.6	0	18	1.0	0.2
5	0.8	0	19	1.4	0.4
6	1.0	0	20	1.8	0.2
7	1.2	0	21	2.2	0.2
8	1.4	0	22	2.6	0.2
9	1.6	0	23	0.4	0.4
10	1.8	0	24	2.4	0.4
11	2.0	0	25	0.8	0.6
12	2.2	0	26	2.0	0.6
13	2.4	0	27	1.2	0.8
14	2.6	0	28	1.6	0.8

Table 4. Element connectivity of the 54-bar truss

Bar	Node	Node	Bar	Node	Node	Bar	Node	Node
1	1	2	19	3	17	37	13	21
2	2	3	20	4	17	38	13	24
3	3	4	21	5	17	39	13	22
4	4	5	22	5	25	40	14	22
5	5	6	23	5	18	41	15	22
6	6	7	24	6	18	42	16	23
7	7	8	25	7	18	43	17	23
8	8	9	26	7	27	44	18	25
9	9	10	27	7	19	45	19	27
10	10	11	28	8	19	46	19	28
11	11	12	29	9	19	47	20	26
12	12	13	30	9	28	48	21	24
13	13	14	31	9	20	49	22	24
14	14	15	32	10	20	50	23	25
15	1	16	33	11	20	51	24	26
16	2	16	34	11	26	52	25	27
17	3	16	35	11	21	53	26	28
18	3	23	36	12	21	54	27	28

Table 5. Two damage scenarios for the 21-bar truss

Scenarios	Damage Bar (s)	Severity of Damage
The first scenario	Bar 4	35%
The second scenario	Bar 13	35%
	Bar 21	25%

Table 6. The damage scenario for the 54-bar truss

Scenario	Damage Bar (s)	Severity of Damage
The first scenario	Bar 30	35%
	Bar 47	25%
	Bar 52	20%

Last but not least, these results lead to the conclusion that early damage identification is critical to maintaining the safety and integrity of structures and reducing maintenance costs. Furthermore, it necessitates improved design, safety, and optimization of complex engineering systems where numerous interacting physical processes occur.

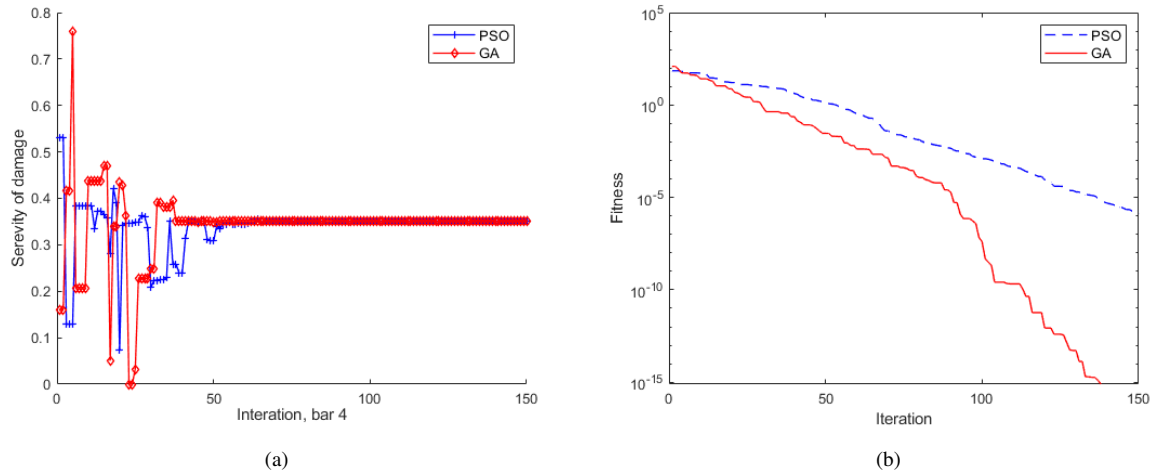


Figure 6. Convergence depiction for the first damage scenario

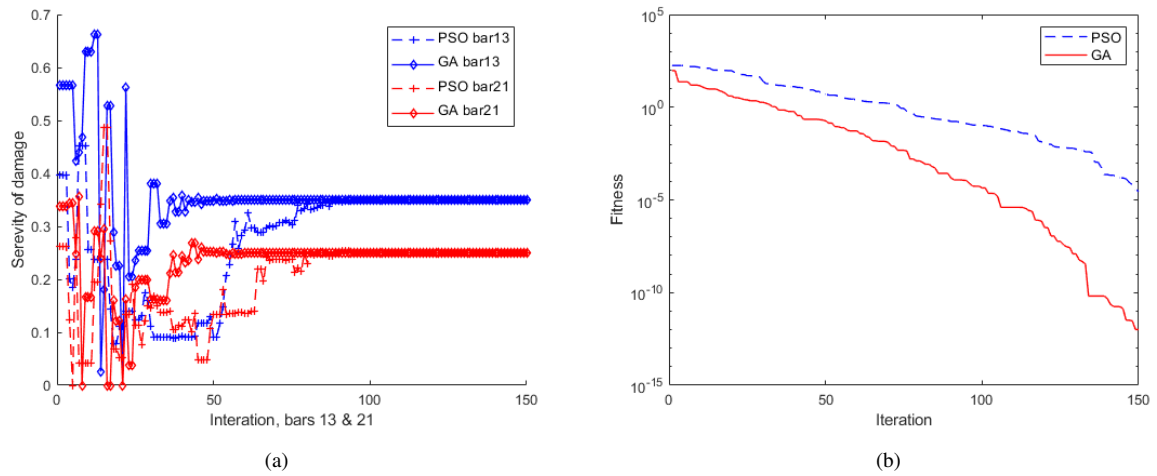


Figure 7. Convergence depiction for the second damage scenario

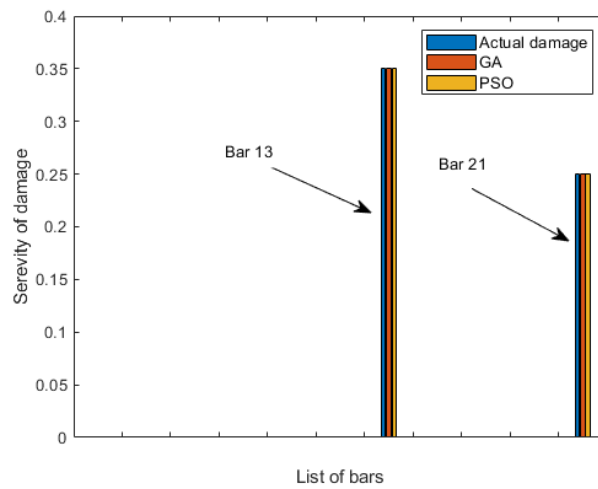


Figure 8. Verification with actual damage for the second scenario

Note: PSO = Particle Swarm Optimization; GA = Genetic Algorithm.

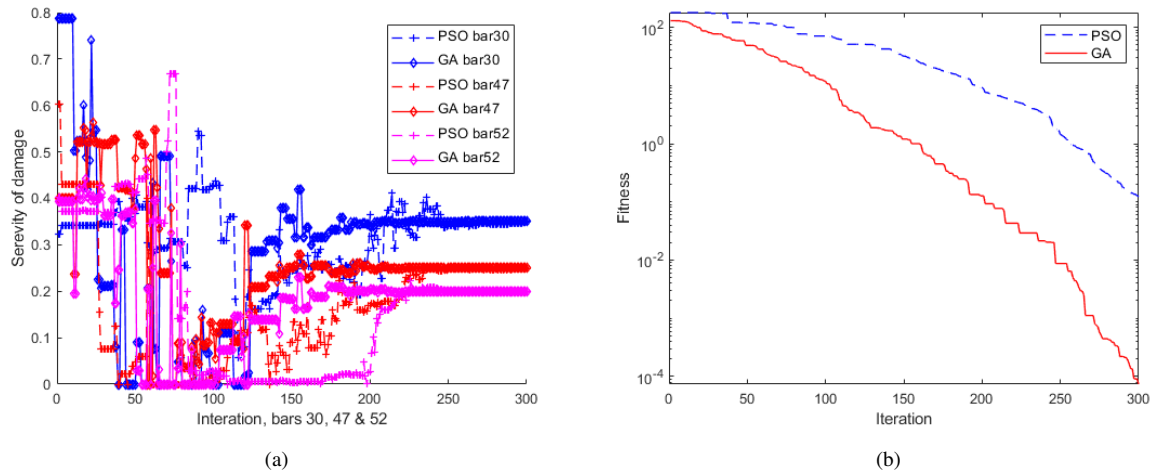


Figure 9. Convergence depiction for the damage scenario of the 54-bar truss

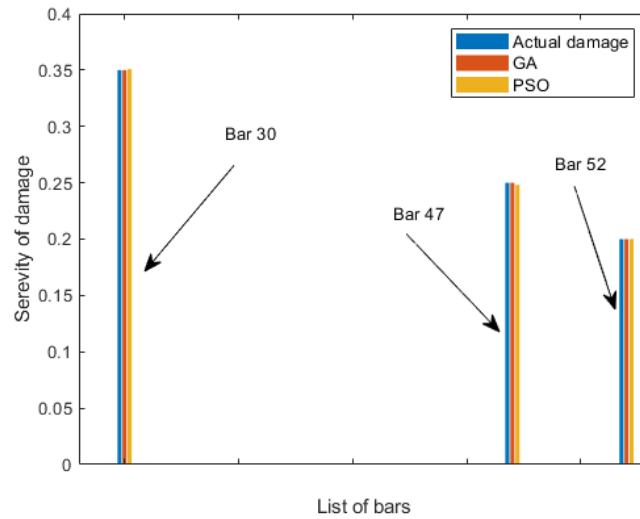


Figure 10. Verification with actual damage for the 54-bar truss

4 Conclusions

In this study, the GA and the PSO were used for damage detection in planar truss structures. A set of scenarios, from simple to complex, was proposed to assess the performance of the optimization algorithms. The obtained results proved that these algorithms not only provided good values for the objective function but also identified the location and severity of damage accurately. Consequently, the demonstrated accuracy of the GA and PSO suggests that these algorithms can serve as a valuable foundation for predictive maintenance and robust design optimization in systems governed by coupled mechanisms and multi-scale effects within a multiphysics environment.

Data Availability

The data used to support the findings of this study are available from the corresponding author upon request.

Conflicts of Interest

The author declares no conflict of interest.

Declaration on the Use of Generative AI and AI-assisted Technologies

The author declares that generative AI or AI-assisted technologies were used only for language editing during manuscript preparation. All scientific content, interpretations, and conclusions were developed and verified by the author, who takes full responsibility for the manuscript.

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